

LESSON 6-4 • SESSION 2 • PRACTICE • FORM A**Order of Operations with Powers**

Learning target – I can evaluate an expression that contains grouping symbols and powers using the order of operations. · 6.EE.A.2c

REMEMBER

- Order: **1.** grouping () **2.** powers **3.** $\times$ and $\div$ left to right **4.** $+$ and $-$ left to right.
- $\times$ and $\div$ are partners: do whichever comes first from the left. So are $+$ and $-$.
- A power applies only to the number just before it: $2 \times 3^2 = 2 \times 9 = 18$.

WORKED EXAMPLE

Evaluate $50 - (1 + 2)^2 \times 4$.

Step 1 Grouping: $1 + 2 = 3$, so $50 - 3^2 \times 4$.

Step 2 Power: $3^2 = 9$, so $50 - 9 \times 4$.

Step 3 Multiply, then subtract: $50 - 36 = 14$.

Answer 14

WORDS order of operations (*orden de las operaciones*) · grouping symbols (*símbolos de agrupación*) · left to right (*de izquierda a derecha*)

PRACTICE

show your work · label units

1. Evaluate each expression.

a. $20 - 3 \times 4 =$ _____

b. $(20 - 3) \times 4 =$ _____

c. $4^2 - 6 \div 2 =$ _____

d. $2 \times 3^3 =$ _____

e. $30 \div 5 \times 2 =$ _____

f. $18 - 10 + 3 =$ _____

2. Two Students. Two students evaluate $8 + 4 \times 3^2 - 10$. One works left to right after the power. The other follows the order of operations.

a. Do the power first: $3^2 =$ _____

b. Multiply next: $4 \times$ _____ = _____

c. Add and subtract left to right. Value: _____

d. Why does working only left to right give a different value? _____

3. Parentheses Power. Evaluate $(4 + 2)^2 \div 4 + 13$.

Hint Grouping first, then the power.

Show your work.

Value _____

4. Four Steps. Evaluate $5^2 + 8 \times (10 - 4) \div 2$.

Hint Show one step on each line.

Show your work.

Value _____

5. Dara's Claim. Dara says $2 + 3 \times 4^2$ has a value of 80. Justify or correct her answer.

Hint Power, then multiply, then add.

Show your work.

Value _____

- 6. Place the Parentheses.** The value of $24 \div 4 + 2 \times 3$ is 12. Insert ONE pair of parentheses so the value changes. Evaluate your new expression.

Hint Not every grouping changes the value. Test yours.

Show your work.

New value _____

- 7. Kwame's Claim.** Kwame says $4 \times 3^2 + 6$ has a value of 150. Justify or correct his answer.

Hint Did Kwame square before he multiplied?

Show your work.

Value _____

- 8. Two Readings.** Use the expressions $4 + 2 \times 6$ and $(4 + 2) \times 6$.

- a. $4 + 2 \times 6 =$ _____
 b. $(4 + 2) \times 6 =$ _____
 c. What did the parentheses change? _____

FIND THE ERROR

explain the mistake, then fix it

- 9. Marcus's Work.** Marcus evaluated $10 + 2 \times 5^2$. He multiplied 2×5 first, then squared that result, and got 110. Explain his mistake. Then find the correct value.

TEST PRACTICE

choose the best answer

- 10.** Find $(2 + 3) \times 4$.

A 14 **B** 20 **C** 24 **D** 9

- 11.** In the order of operations, powers come –

A before $\times$ and $\div$ **B** after $+$ and $-$
C last **D** before grouping symbols

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

22 O	34 P	150 X	98 Q	16 S	49 W	42 R	50 E	110 J	80 Z
		#2c	#3	#4	#5	#7	#8a		

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____